REVISION A · OFFLINE BUILD WORKSHEET

FRC Sensor CAN setup

Choose your sensor outputs, check the shared power budget, and record the jumpers and service command for this board.

Channel 1

J4

Sensor output

Industrial NPN · battery power

Three-wire sensor. Port pull-up; sensor must work over the actual battery range.

Sensor maximum supply current, mA

25

- [x] Logical TRUE when the output is LOW

Channel 2

J5

Sensor output

Industrial PNP · battery power

Three-wire sensor. Pulldown; sensor must work over the actual battery range.

Sensor maximum supply current, mA

35

- [ ] Logical TRUE when the output is LOW

Channel 3

J6

Sensor output

5 V push-pull · 5 V power

A driven logic output. No bias shunt. Sensor must accept regulated 5 V power.

Sensor maximum supply current, mA

50

- [ ] Logical TRUE when the output is LOW

Channel 4

J7

Sensor output

Dry contact to ground

Contact connects SIGNAL to GND. No sensor power; use the onboard 3.3 V pull-up.

Sensor maximum supply current, mA

0

- [x] Logical TRUE when the output is LOW

Jumper worksheet

Identify pin 1 from the square pad and assembly drawing. These settings refer to numbered pins, regardless of header orientation.

PORT	POWER	THRESHOLD	BIAS
J4	JP101 2–3 / battery	JP102 open / industrial	JP103 1–2
J5	JP201 2–3 / battery	JP202 open / industrial	JP203 5–6
J6	JP301 1–2 / 5 V	JP302 fitted / logic	JP303 open
J7	JP401 open	JP402 fitted / logic	JP403 3–4



Port pull-up · 3.3 V pull-up · Pulldown

Install one shunt at most on each power or bias selector. Disconnect board power before changing jumpers. JP1 is momentary reset; leave it open. JP2 is CAN termination; fit it only at a physical end of the trunk.

Connection limits

- Each sensor connector is pin 1 POWER, pin 2 GND, pin 3 SIGNAL. The board targets 0–30 V positive signals and a 6–18 V protected board supply.
- Battery power follows the robot battery. A sensor requiring 10 V can fail during a battery sag while this interface continues running.
- Logic mode targets LOW ≤0.8 V and HIGH ≥2.4 V. Industrial mode targets LOW ≤4 V and HIGH ≥7 V.
- Industrial NPN/PNP examples assume ≤0.5 mA off leakage. The 3.3 V open-drain pull-up example assumes ≤50 µA. Check the actual sensor datasheet.
- External supply voltage must never enter a POWER pin. Externally powered 24 V sensors need the wiring arrangements in the full guide.

This worksheet performs arithmetic and selects documented modes. It does not establish sensor compatibility, measured electrical ratings, or robot inspection approval.

Full wiring instructions · Printable handbook · Interactive assembly BOM

Shared rail budget

Regulated 5 V **50.1 / 300 mA**

Battery **68.0 / 300 mA**

Sum of entered sensor currents and conservative bias-load allowances. The 300 mA budget applies to each complete rail, shared by all four ports. Startup current requires a separate bench check.

> Entered steady currents fit both rail budgets. Verify the real sensors and startup current on the bench.

Board configuration

Device ID

17

Debounce, ms

2

Service serial port

/dev/ttyUSB0

- [x] Save selected sensor rails as ON at startup, after wiring and rail tests pass.

Inversion mask 9. Selected rails 3 (1=5 V, 2=battery). Saved startup rails 3.

Provisioning command

```
python host/sensor_can.py configure --port /dev/ttyUSB0 --device 17 --invert 9 --debounce 2 --rails 3
```

Run from the project folder using the 3.3 V UART service connection. This page never connects to or programs the board.

Before enabling power

Verify polarity, shunts and both rail voltages with a current-limited supply. Assign a unique ID before connecting to the robot CAN bus. Use the supplied robot driver’s stale-data checks and explicit fault recovery.

Original sensor interface · Hardware A · Firmware 0.1.0 · Protocol 1
Dated design worksheet: 2026-09-07